Chapter 9: Evaluation Techniques

9.1 What is Evaluation?

- to assess our designs and test our systems to ensure they behave as expected and meet user requirements.
- helps determine if users can use the system effectively.
- used to find problems before the final product is completed
- Accesses usability and functionality of an interactive system
- Evaluation is not a single phase / final step
- Continuous process that should occur throughout the design process
- Results of evaluation should be used to improve the design.

Why evaluate early:

- much easier (and cheaper) to change a design in the early stages
- Problems can be fixed before spending time and money on implementation (large effort n resources)
- Prototypes and design principles help identify issues early.

Two Main Types of Evaluation

1. Evaluation by Designers/Experts OR Expert-based Evaluation

- No real users involved.
- Done by usability experts or designers.
- Useful for early designs and prototypes.
- Quick cost effective feedback

2. Evaluation with Users / User-based Evaluation

- Real users interact with system.
- Usually requires a working prototype or completed system.
- Helps understand actual user behavior.

The distinction is not strict:

- Users may participate early through focus groups.
- Experts can also evaluate finished systems.

9.2 Goals of Evaluation

Goal 1: Assess Functionality and Accessibility

system should:

- Match user requirements.
- Provide needed functions.
- Make functions easy to find / easy to access
- Allow users to perform tasks efficiently.
- Meet user expectations.
- Measures speed, accuracy and effectiveness
- Evaluation at this level may also measure the user's performance to assess the system's effectiveness in supporting the task.

Example : If clerks normally search customer records by address, then a computerized system should also allow address-based searching.

Goal 2: Assess User Experience

Evaluation checks:

- Ease of learning
- Ease of use
- User satisfaction
- Enjoyment
- Emotional response

It is important to identify areas that overload the user [Mental workload + Memory burden + Confusion + Frustration]

Example: If users must remember too many commands, the system overloads memory.

Goal 3: Identify Design Problems

finds:

- Confusing features
- Unexpected behaviors
- Errors
- Difficult interactions

Purpose:

- Locate trouble spots.
- Fix them.

9.3 Evaluation Through Expert Analysis

Evaluation should ideally happen before implementation.

Benefits:

- Avoids expensive mistakes.
- Saves time and resources.
- Allows design improvements early before major resource commitments
- The later an error is discovered, the more costly it is to fix

User testing:

- Costs money.
- Takes time.
- Requires working prototypes.

Expert analysis:

- Is cheaper.

- Can be used on incomplete designs
- Flexible (usable from/can evaluate design specification, through storyboards and prototypes, to full implementations)

The aim is to identify areas likely to cause user difficulties & usability issues because they violate known cognitive principles or ignore accepted empirical results.

Properties of expert methods: flexible

Limitation:

- Does not study actual user behavior.
- Only predicts likely usability problems.

Four approaches are considered

1. Cognitive Walkthrough
2. Heuristic Evaluation
3. Model-Based Evaluation
4. Previous Studies

9.3.1 Cognitive Walkthrough

- A method where experts go through each step required to complete a task
- Purpose:
    - *Find* usability problems.
    - Evaluate ease of learning thru exploration — because many users prefer to learn hands-on rather than through training or manuals.

In a cognitive walkthrough, the sequence is the steps an interface requires a user to perform to accomplish a known task; evaluators 'step through' that sequence to check for potential usability problems.

Four things you need to do a walkthrough

1. A system specification or prototype (need not be complete, but fairly detailed — even menu location and wording/labels matter)

2. A description of the task the user is to perform (realistic task most users will want to do)
3. A complete, written list of the actions needed to complete the task [action sequence] with the proposed system.
4. User description - what experience/knowledge/skill level the evaluators can assume about them.

Four questions asked for each action

1. **Is the effect of the action the same as the user's goal at that point? / will user try to achieve the correct goal?** (e.g., if the action saves a document, is 'saving' what the user wants?)

2. **Will users see that the (correct) action is available?** Is the button/menu item visible at the time it is needed? this is about visibility & accessibility,

3. **Will users understand the action? / Once users have found the correct action, will they know it is the one they need?** This is about whether its meaning and effect are clear (meaning of labels/icons) ⇒ recognition

4. **Will users understand the feedback they get?** Sufficient confirmation of what happened (Confirmation messages, System responses) — this completes the execution–evaluation cycle ; users *need appropriate feedback to know they achieved their goal.*

Documentation

- everything should be recorded.
- vital to document the walkthrough to record what is good and what needs improvement

› Forms Used

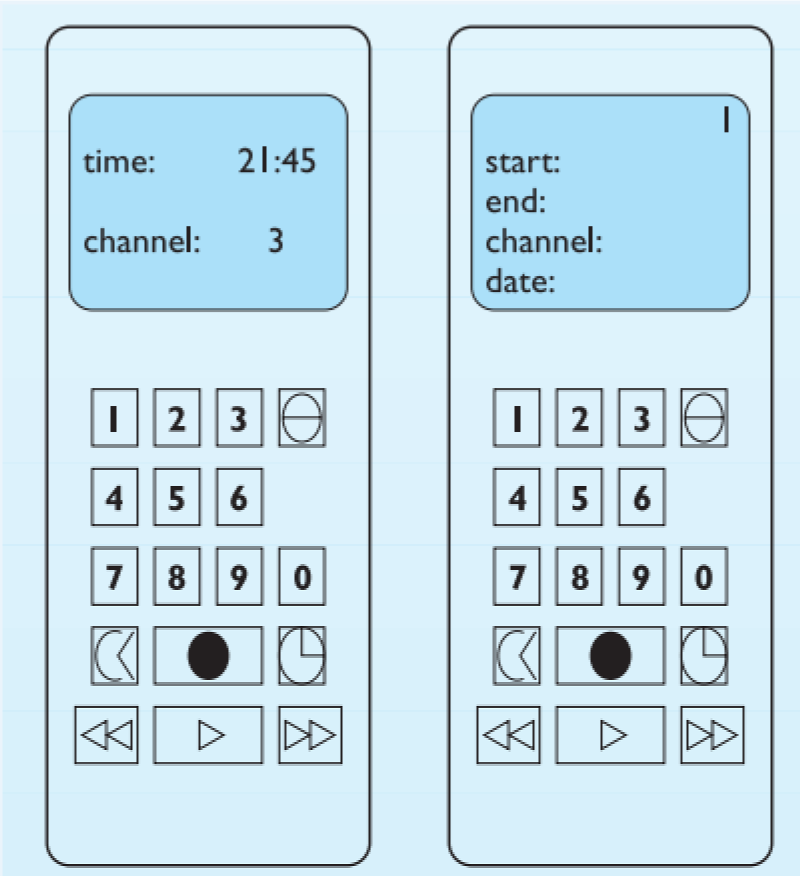
- **cover form**
  - System
  - Task
  - Users

    - Evaluators
    - Date/time
- **Action form**
    - separate form per action (answering the four questions).
- **Usability Problem Report**
    - Used when a problem is found
    - Records the system/version, date, evaluators, and a detailed problem description.
    - It is useful to note severity (how often the problem occurs and how serious it is for users) → helps prioritize fixes

Example — programming a VCR by remote control

Task: program the video to time-record a program **starting 18.00**, **finishing 19.15, channel 4**, on **24 February 2005**. Assume the user is familiar with VCRs but not this design. The action sequence is written as alternating User Action (UA) and System Display/response (SD).

- Start: 18:00
- End: 19:15
- Channel: 4
- Date: 24/02/2005



UA1: press 'timed record' → SD1: display moves to timer mode; flashing cursor after 'start:'

UA2: press 1 8 0 0 → SD2: digits shown; cursor moves on

UA3: 'timed record' → SD3: cursor to 'end:' • UA4: 1 9 1 5 → digits shown

UA5: 'timed record' → cursor to 'channel:' • UA6: 4 → digit shown

UA7: 'timed record' → cursor to 'date:' • UA8: 2 4 0 2 0 5 → digits shown

UA9: 'timed record' → stream number flashes • UA10: 'transmit' → details sent, display returns to normal

Walkthrough of UA1: Q1 yes (timed record initiates programming, a reasonable first goal). Q2 yes (button is visible). Q3 — a problem: it is not clear WHICH button is 'timed record' — the clock icon is the correct choice but could be read as 'change time'; the filled circle (record) or another button are rival candidates, so the user may fail here. Q4 yes (familiar headings start/end/channel/date confirm success). The identified usability problem concerns the icon used on the 'timed record' button.

9.3.2 Heuristic Evaluation

- heuristic is a guideline, general principle or rule of thumb used to guide or critique a design decision.
- developed by Jakob Nielsen and Rolf Molich
- Experts inspect a design using predefined usability rules
- Fast, flexible and cheap, hence often called a discount usability technique

› **PROCESS**

- several evaluators independently critique/inspect the system to find potential usability problems.
- AVOIDS BIAS
- FINDS MORE PROBLEMS
- Nielsen's experience: 3–5 evaluators recommended
- five typically finding about 75% of usability problems.

Severity rating (each problem rated 0–4)

Severity combines four factors: how common the problem is, how easily the user can overcome it, whether it is a one-off or persistent, and how seriously it is perceived.

Rating	Meaning
0	Not a usability problem.
1	Cosmetic problem only — fix only if extra time is available
2	Minor usability problem — low priority
3	Major usability problem — important to fix, high priority
4	Usability catastrophe — must fix before release

Nielsen's ten heuristics

Heuristic	Description	HCI Example

1. Visibility of System Status	Always keep users informed about what is happening through timely and appropriate feedback.	Displaying a file upload progress bar showing percentage completed.
2. Match Between System and Real World	Use language, concepts, and terminology familiar to users instead of technical jargon.	Labeling a checkout button "Add to Cart" instead of "Execute transaction_id".
3. User Control and Freedom	Allow users to easily undo, redo, or cancel actions when mistakes occur.	Gmail's "Undo Send" option after sending an email.
4. Consistency and Standards	Follow established conventions so users do not have to learn different meanings for similar actions.	Placing the Settings icon in the top-right corner as commonly seen in applications.
5. Error Prevention	Prevent errors before they happen rather than only displaying error messages afterward.	Disabling the Submit button until all required fields are correctly completed.
6. Recognition Rather Than Recall	Make options, actions, and information visible so users do not need to remember them.	Displaying recent searches or search suggestions in Google Search.
7. Flexibility and Efficiency of Use	Provide shortcuts and accelerators that improve efficiency for experienced users while remaining easy for beginners.	Supporting keyboard shortcuts such as Ctrl + C and Ctrl + V.

8. Aesthetic and Minimalist Design	Keep interfaces simple and uncluttered by removing unnecessary information.	The clean and minimalist homepage of Google Search.
9. Help Users Recognize, Diagnose, and Recover from Errors	Error messages should clearly explain the problem and suggest how to fix it.	"Incorrect Password. Click 'Reset Password' to recover your account."
10. Help and Documentation	Provide easily accessible help resources with clear instructions and task-oriented guidance.	An application's Help Center with a searchable knowledge base.

Once each evaluator finishes their separate assessment, all problems are collected, mean severity ratings calculated, and the design team decides which to address first.

9.3.3 Model-based Evaluation

Model-based evaluation uses mathematical, cognitive, or design models to predict how users will interact with a system.

- Instead of observing users directly, we use models to predict usability
- Helps compare design alternatives before implementation.
- Useful in early design stages.

1) GOMS (Goals, Operators, Methods, Selection)

- Goals : What the user wants to achieve.
- Operators: Basic actions performed by the user.
- Methods: Different ways to achieve a goal.
- Selection Rules: Decide which method to use.

> Purpose

- Predict user performance.

- Compare interface designs.
- Estimate efficiency.

2) Keystroke-level model KLM

- Detailed version GOMS
- **Predicts how long users take to perform tasks**
- Very accurate for simple tasks.
- Helps compare designs quickly.
- Measures
  - i) Keystrokes
  - ii) Mouse movements
  - iii) Mouse clicks
  - iv) Mental preparation time

3) Design rationale

- **method for recording design decisions**
- design options are evaluated by examining the criteria associated with each option and the supporting evidence, enabling informed judgments.

4) Dialog models

- Used to analyze system dialogs
- evaluate dialog sequences for problems such as unreachable states (a screen/state user can never access), circular dialogs (user gets trapped in loops) and complexity (interaction becomes unnecessary complicated)
- prior to implementation → problems can be found before coding
- e.g., state transition networks

9.3.4 Using Previous Studies in Evaluation

Instead of conducting new experiments every time:

- Use existing research findings.
- Apply previous HCI results.

Previous studies may already tell us:

- Which menu type works best.
- Which icons are easiest to understand.
- How many items a menu should contain.
- Whether users remember commands better than icons.

> saves time + memory
> evidence based decision
> experts involved !!!

9.4 Evaluation Through User Participation

Expert reviews are useful, but real users are essential.

Reason:

- **Experts predict problems.**
- **Users reveal actual problems.**

User participation usually occurs in **later development stages** when at least a **working prototype exists**

User-participation approaches include

1. **Experimental Methods** → Controlled experiments.
2. **Observational Methods** → Watching users.
3. **Query Techniques** → Asking users questions.
4. **Physiological Monitoring** → Measuring physical reactions.

9.4.1 Styles of Evaluation

1) **Laboratory Studies**
2) **Field Studies**

1) Laboratory studies

Users are **tested in a controlled environment** → **usability laboratory/quiet room**

> **BENEFITS:**

- **Better recording** ⇒ Using cameras, mics, eye trackers
- **Fewer interruptions**
- **More control** ⇒ Researchers control conditions.

> **Disadvantages:**

- **lack of real context** (filing cabinets, calendars, interruptions)
- **Artificial environment** → users may behave differently

> **Suitable Situations**

- **Dangerous/remote locations** → (e.g., a space station)
- **Controlled comparisons** ⇒ Comparing two interface designs fairly.
- **Rare situations** ⇒ testing uncommon effects

2) Field studies

Evaluation takes place in the **user's real environment**. [workplace]

> **advantages:**

- **Realistic** ⇒ users behave naturally
- **Context preserved** ⇒ users natural env, interruptions, communication, habits
- **Long-term activities** ⇒ Can study tasks lasting: Days, Weeks or Months

> **Disadvantages:**

- **Hard to observe** — noise, movement and constant interruptions
- **Less control**

Heisenberg Effect

- People change behavior when being observed.
- Even field studies cannot completely eliminate this effect.

field observation is generally **preferred** since it studies interaction in actual use, **Laboratory Studies** / Controlled experiments, however, are useful for evaluating specific interface features and normally need lab conditions.

9.4.2 Empirical Methods: Experimental Evaluation

A **controlled experiment** used to **test hypotheses.**

Purpose : Provide objective evidence.

Basic Process

1. Create hypothesis.
2. Manipulate variables.
3. Measure results.
4. Analyze statistically

⭐Participants

- Participants should resemble real users.
- If not actual users, match similar age, education, general computer experience, experience with related systems, and task-domain knowledge.
- **Sample size:** must be large enough to be representative, given the experimental design and statistical methods
- **Observational studies** ⇒ 5 users
- **Statistical experiments** ⇒ at least 10 or more

⭐Variables

- **Independent variable (IV)**
    - **manipulated to produce conditions for comparison**
    - The factor being **changed**
    - (e.g., interface style, level of help, number of menu items, icon design).

  Levels → different values of IV

    - Each value used is a level (e.g., **menus of 5, 7, 10 items = three levels**)
    - Experiments may have more than one IV — two IVs with 3 and 2 levels need **3 × 2 = six conditions**

- **Dependent variable (DV)**
    - **what is measured**
    - **value depends on changes to the IV**
    - (e.g., speed of menu selection)
    - It must be measurable, affected by the IV, and minimally affected by other factors.
    - Common DVs: time to complete a task, number of errors, user preference, quality of performance.

⭐Hypotheses

- **Prediction about results**
- predicts the experiment's outcome, framed **in terms of IV and DV**
- **null hypothesis** —states there is **no difference** in the DV between IV levels.
- Researchers try to reject the null hypothesis
- Statistical measures are compared against **levels of significance**;
- **Statistical Significance:** Shows whether differences are real or happened by chance. [95% confidence means only 5% chance the result occurred randomly]

⭐Experimental design

First choose the hypothesis (deciding exactly what to demonstrate); this clarifies the IV and DV. Consider participant availability and representativeness. Then choose the experimental method:

3 major designs:

1) **Between-Subjects Design**
2) **Within-Subjects Design**
3) **Mixed Design**

- **Between-subjects (randomized)**
  - each participant is assigned to one condition
  - **Advantage:** no learning/transfer effect (each user does only one condition).
  - **Disadvantages**: more participants needed; between-group variation and individual differences can **bias/negate results**

- **Within-subjects (repeated measures)**
  - Every participant performs under every condition.
  - **Disadvantages**: suffer transfer of learning, reduced by varying the order (group A: 1 then 2; group B: 2 then 1)
  - **Advantages**: Less costly (fewer users), Good when learning varies and less affected by variation between participants.

- **Mixed design**
  - Combination of between-subjects and within-subjects.
  - a popular compromise with more than one IV — one variable between-groups, one within-groups

Finally, decide how to analyze the results — the choice of statistical test is vital, since different tests assume different things and an inappropriate test can invalidate the results.

⭐Statistical measures

First two rules:

1) **look at the data** ⇒ graphs/tables/look for unusual values → visual inspection reveals problems
2) **save the raw data** ⇒ keep original data ⇒ re-analyze later / verify results / find errors

A glance at a graph/histogram/table is often more instructive than a blind test and **exposes outliers** (a value very different from the rest), outliers can distort averages and statistical results.

› Types of Variables

- **Discrete variable** — takes only a finite number of values/levels (e.g., screen color red/green/blue). In interface experiments the **IV is usually discrete**.
- **Continuous variable** — can take any value within a range. E.g. height/weight/task time

› Parametric tests

- Statistical tests that assume data follows a known distribution → **Normal Distribution (Bell Curve)**
- More powerful, sensitive.
- Can detect small differences
- Examples:
  - Student's t-test → used when comparing 2 groups
  - ANOVA → Comparing more than two groups
  - Regression → Studying relationships between variables

› Non-Parametric tests

- Do not assume any distribution.
- **Use rankings**.
- More resistant to outliers
- Handles unusual data
- Less powerful, miss small differences

› Contingency Tables

- Tables used when variables are categorical.
- Used with: **Chi-Squared Test**
  - Determines whether categories are related.

Table — choosing a statistical technique:

Independent variable	Dependent variable	Test
Parametric — Two-valued	Normal	Student's t test on difference of means
Parametric — Discrete	Normal	ANOVA (Analysis of Variance)
Parametric — Continuous	Normal	Linear (or non-linear) regression / factor analysis
Non-parametric — Two-valued	Continuous	Wilcoxon (or Mann-Whitney) rank-sum test
Non-parametric — Discrete	Continuous	Rank-sum versions of ANOVA
Non-parametric — Continuous	Continuous	Spearman's rank correlation
Contingency — Two-valued	Discrete	No special test (see next entry)
Contingency — Discrete	Discrete	Contingency table and chi-squared test
Contingency — Continuous	Discrete	(Rare) group the independent variable, then as above

Three questions statistics can answer

- **Is there a difference?** hypothesis testing — answers are probabilistic, e.g., 'we are 99% certain selection from 5-item menus is faster than from 7-item menus.'
- **How big is the difference?** point estimation (often averages), e.g., 'selection from 5 items is 260 ms faster.'
- **How accurate is the estimate?** Confidence interval — measures of variation (standard deviation) or confidence intervals, e.g., 'faster by 260 ± 30 ms; 95% certain the difference is between 230 and 290 ms.

[didnt do this poori idk kia hai krni hai ya nhi]

Worked example — non-parametric (Wilcoxon) statistics

Data — **condition A: 33, 42, 25, 79, 52; condition B: 87, 65, 92, 93, 91, 55**. Pool and rank all values; A becomes ranks 2,3,1,7,4 and B becomes 8,6,10,11,9,5. For each condition, sum the ranks and subtract the least possible sum (15 for A's five values, 21 for B's six). This gives U = 2 (A) and U = 28 (B); their sum (30) equals 5 × 6, so only one U need be computed. Compare the smaller U with critical values (here, for 5 and 6 values, the 5% critical value is 3). Since 2 < 3, reject the null hypothesis — there is likely a real difference (only a 1-in-20 chance it happened by chance). The test was correct: the data were random 1–100 with 10 subtracted from condition A.

This example uses the **Wilcoxon Rank-Sum Test (Mann–Whitney U Test)**, which is a **non-parametric test**. It is used when we want to compare two independent groups without assuming the data follows a normal distribution.

The data is:

- Condition A: 33, 42, 25, 79, 52
- Condition B: 87, 65, 92, 93, 91, 55

First, combine all values and sort them from smallest to largest:

Value	Rank
25	1
33	2
42	3
52	4
55	5
65	6
79	7
87	8
91	9
92	10
93	11

assign ranks to each group: Ranks of A = **2, 3, 1, 7, 4**, Ranks of B = **8, 6, 10, 11, 9, 5**

Sum the ranks: RA=2+3+1+7+4=17, RB=8+6+10+11+9+5=49

Worked exercise — testing color coding for accuracy

Hypothesis: color coding will make selection more accurate. Participants: from the user population. IV: color coding. DV: accuracy (number of errors). Design: between-groups to avoid transfer of learning (or within-groups with safeguards if participants are scarce). Task: identical interfaces except the second adds color to indicate related menu items; the participant is told verbally what to select and must select within a strict time limit before the screen clears; failure = an error; item positions change each presentation. Analysis: t test.

› Studies of groups of users

Evaluating group systems (groupware) is harder than single-user experiments because of the complexity of human–human communication.

- **Participant groups** — participant requirements; 10 group-of-three experiments need 30 participants; group sessions run longer (the group must 'settle down' and build rapport) → more disruption/expense. Scheduling many people + equipment is hard. Many reports use only three or four groups — a statistical problem, but not the primary obstacle.
- **Data Collection Complexity -** Need to collect: Multiple data like videos, audio streams, logs simultaneously
- **Large Group Differences -** One group may Cooperate perfectly while another may constantly argue

Solutions to Small Group Numbers

- **Within-Group Experiments** → same group experiences multiple conditions - reduces group differences
- **micro-analysis** → Analyze tiny details - eg; time b/w responses, length of pauses
- **anecdotal analysis** → Focus on important incidents. (communication breakdown, major successes)

› Field Studies with Groups

Group work is highly dependent on context.

- Problems:
  - **Artificial Groups** ⇒ Randomly created groups behave unnaturally.
  - **Removing Existing Groups** ⇒ Changes their normal behavior.
  - **New Systems** ⇒ May not yet have a normal working environment.

Ethnography

Detailed study of people in their natural environment.

Researcher:

- Observes
- Records
- Does not interfere

Provides deep understanding of:

- Communication
- Cooperation
- Work practices

9.4.3 Observational Techniques

- Observe users performing tasks.
- **Purpose**: Understand actual behavior.

Think aloud and cooperative evaluation

- Users continuously explain what they are thinking while using the system.
- They speak their thoughts aloud during task performance.
- "I think this button will open settings." "I don't know what this icon means." "I'm trying to save the file."
- Allows evaluators to understand: User reasoning, Decision-making process Confusion points, Expectations
- **Advantages**: simple, little expertise needed, gives useful insight, usable throughout design

- **Disadvantages:** subjective - diff users explain differently, task dependent, Can Affect Behavior - Talking while working may change how users perform tasks.

› **Cooperative evaluation**

- Modified version of think aloud
- user is a collaborator, not just a participant.
- User and evaluator work together.
- **User Can:** Ask questions, Give opinions, Criticize the design
- **Evaluator Can:** Ask follow-up questions, Request clarification
- **Advantages**: less constrained and easier for evaluator to learn; the user is encouraged to criticize the system; the evaluator can clarify confusion as it occurs, maximizing effectiveness at identifying problem areas.

The record of such a session is called a **protocol [** complete record of an observation session **]**

Protocol analysis — methods of recording

- **Paper and pencil** — **evaluator manually writes observations**.
  - Cheap, flexible
  - Slow, easy to miss events - fast actions may not be recorded
  - Can reduce writing time if we use codes (E - error)
- **Audio recording** — **useful when the user thinks aloud**, but may not capture enough to identify exact actions later, and is hard to match to another protocol form (e.g., a handwritten script).
- **Video recording** — **Record user and system interaction**
  - shows what the participant is doing (if they stay in range).
  - Events can be reviewed repeatedly.
  - Choosing camera positions/angles for enough detail while keeping the participant in view is difficult (asking them not to move is unnatural).
  - Users may behave differently when filmed.
- **Computer logging** **—System automatically records user actions.**
  - logs: clicks, keystrokes, commands, menu selections
  - automatic , user doesnt notice it so theyre not interrupted
  - Good for long-term studies - can run for days - months
  - Huge data volume - thousands of actions may be recorded

    - No context - shows what happened not why
- **User notebooks — participants keep their own activity/problem logs**
    - Useful when observation is impossible.
    - Suer may forget details - incomplete
    - Subjective

Usually multiple methods are used together.

Problems in Protocol analysis

- Synchronizing multiple sources during playback is difficult
- Transcription - converting recordings into text
- needing annotation of voices and non-verbal items (pauses, emphases, phones ringing)

Automatic protocol analysis tools

Manual analysis is extremely time-consuming. Special software tools help. Automatic tools edit, annotate and synchronize video/audio/system logs:

- **EVA (Experimental Video Annotator)**
    - Annotates videos during usability testing.
    - Evaluator can mark Errors, Important events, Screenshots, Notes while watching.
    - Evaluator may become distracted while tagging events.
    - Important events may be missed.

- **Workplace project (Xerox PARC)**
    - Analyze multiple synchronized data streams
    - COMBINES video, audio, notes, diagrams in aligned displays, aligned by timestamps or event/action via hypertext links.

- **DRUM — video annotation/tagging within the MUSiC toolkit (Measuring the Usability of Systems in Context)**
    - Measure usability using structured methods.

    - supporting a complete methodology based on usability metrics (analytic metrics, cognitive workload, performance, user satisfaction); DRUM focuses on measuring performance and is supported by manuals, questionnaires, analysis software and databases.

Post-task walkthroughs

- After completing tasks, users review their actions and explain them.
- Direct observation data often lack interpretation
- **transcript** is replayed and the participant comments or is questioned.
- **Immediate Walkthru:**
    - Reasons are still fresh in memory.
    - Users remember exactly what they were thinking.
- **Delayed walkthru**:
    - User memory may fade, but:
    - Evaluator has time to:
        - Analyze data
        - Identify important events
        - Prepare better questions

when the participant cannot talk during observation (a critical or too-intensive task), or when minimizing non-task talk is preferred for natural performance — making the post-task walkthrough essential.

9.4.4 Query Techniques

- directly asking users for information.
- Gather users opinions, preferences, experiences, satisfaction, suggestions
- Direct user feedback
- Users may identify issues designers never considered.
- Easy n inexpensive
- But it is subjective - response depends on personal opinions
- 'rationalized' answers - what they say differs from what they actually did.
- Hard to Evaluate New Designs

Main Types of Query Techniques

1. Interviews
2. Questionnaires

› Interviews

- direct conversation between evaluator and user.
- Used to collect detailed information.
- Top down approach - broad → more detailed qs
- Before conducting interviews **Prepare Core Questions**
    - Ensures: Focus, Consistency, Coverage of important topics
- Flexible, Deep Investigation , Misunderstandings can be corrected immediately, Reveals Unexpected Problems
- Time Consuming , Difficult to Analyze (responses vary), less consistent (Different interviewers may ask questions differently.)

› Questionnaires

- Less flexible than interviews
- written set of questions completed by users.
- Collect information from many users efficiently
- questions fixed in advance, but reach a wider group, take less time to administer, and easy to analyze , consistent (Every user receives identical questions)
- Since the evaluator usually isn't present, the questionnaire must be well designed — first establish its purpose (what information is needed) and how responses will be analyzed, target users

Styles of question:

1. General Questions:
collect basic background information about respondents, such as age, gender, occupation, education, and computer experience. help researchers determine whether the participants match the target user population and ensure that the results are relevant to the intended users.

2. Open-Ended Questions:
allow users to answer in their own words without predefined options. They provide rich, detailed feedback and can uncover unexpected issues or ideas. However, responses are difficult to analyze because they vary greatly, and users may avoid answering if the questions require too much writing. They are best used as supplementary questions alongside other question types.

3. Scalar Questions:
ask users to rate something on a numerical scale, such as 1–5. They are commonly used to measure opinions, satisfaction, or agreement with statements. These questions are easy to analyze statistically, allowing researchers to calculate averages, percentages, and trends, and compare results across different systems. **Odd-numbered** scales include a neutral option, while even-numbered scales force users to choose a positive or negative opinion

4. Multiple-Choice Questions:
set of predefined answers from which users select one or more options. They are quick and easy for users to answer and simple for researchers to analyze because responses can be counted and categorized. Yes/No questions are the simplest form of multiple-choice questions.

5. Ranked Questions:
require users to arrange a list of options in order of preference or importance. They are useful for identifying user priorities and understanding which features, services, or options users value most.

Design and distribution tips: favor closed questions (scalar/ranked/multi-choice) to reduce respondent effort, encourage a high response rate, and ease analysis (percentages, correlations, factor analysis). Always run a **pilot testing** with **4-5 users** to iron out problems before wide distribution. Return rates are low (often 25–30%), so distribute many more questionnaires than needed. Participants should represent the target population.

9.4.5 Monitoring Physiological Responses

- rely on observation and on users telling us what they do/feel.
- Physiological monitoring measures physical responses directly

The two main areas are:

1) **eye tracking**
2) **physiological measurements**

👁️Eye tracking

- Technology that **records where users look.**
- Eye movements reflect the amount of cognitive processing
- Modern eye trackers → head mounted / desk mounted
- Early eye trackers → required devices attached directly to the eye.
- display requires (how easy/hard it is to process), so where people look and their movement patterns indicate which screen areas are easy/difficult.
- Measurements are based on **fixations** (the eye remains focused on one location) and **saccades** (rapid ballistic movement between points). Usability measures include:
  - **Number of fixations** — more fixations → less efficient search strategy + confusion
  - **Fixation duration —** longer fixations may indicate difficulty with a display.
  - **Scan path —Complete route followed by the eyes.** indicates areas of interest, search strategy and cognitive load; the **optimal//ideal** scan path goes straight to target with short fixation there.

Advantages of eye tracking ⇒ objective - reveals hidden patterns - valuable for website design

Limitations ⇒ expensive - hard to interpret

🧠Physiological measurements

- Measure **bodily responses during interaction.**
- Emotions affect the body

Probes/sensors measure:

- **Heart activity** — blood pressure, volume, pulse — may respond to stress or anger.
- **Sweat-gland activity - galvanic skin response (GSR)** — thought to indicate arousal and mental effort
- **Muscle electrical activity (EMG - Electromyography)** — appears to reflect physical involvement in a task, effort, engagement
- **Brain electrical activity (EEG)** — associated with decision making, attention and motivation.

Open research questions: the relationship between interaction events and these measures is unclear — e.g., does increased pulse indicate frustration or stress at not completing the task? How do responses differ for discrete events vs continuous use? Can patterns map to specific emotional states?

9.5 Choosing an Evaluation Method

- **no fixed rules** for choosing an evaluation method.

9.5.1 Factors Distinguishing Evaluation Techniques

At least eight factors help make an appropriate choice:

1. Design vs Implementation:
At what stage is the system? During the **design stage**, when only sketches, mockups, or specifications exist, methods like **Cognitive Walkthrough, Heuristic Evaluation,** and **Modeling Techniques** are preferred because they are quick, inexpensive, and do not require users. During the **implementation stage,** when a prototype or working system exists, methods such as **experiments, observations, interviews, and questionnaires** can be used because users can interact with the actual system.

2. Laboratory vs Field Studies:

Laboratory studies are conducted in a controlled environment, making observation, recording, and control easier, but the setting may feel artificial and less realistic. **Field studies** are conducted in the users' actual work environment, providing more natural behavior and realistic context, but they are harder to control and observe. A combination of both is often ideal, with laboratory studies used early in development and field studies used later with the real system.

3. Subjective vs Objective Techniques:

Subjective evaluation relies on human judgment and opinions, such as Think Aloud sessions, Cognitive Walkthroughs, and interviews. These methods provide rich insights but may be affected by evaluator bias, which can be reduced by using multiple evaluators. **Objective evaluation** uses measurable data such as controlled experiments, eye tracking, and physiological measurements. These methods are more scientific, repeatable, and easier to compare, but they may overlook unexpected usability issues. Using both approaches together provides the most complete evaluation.

4. Qualitative vs Quantitative Measures:

Quantitative measures produce numerical data such as task completion time, error rates, and success rates, making statistical analysis and comparisons easy. Qualitative measures provide descriptive data such as user comments, opinions, and observations, helping explain why problems occur. Although qualitative data is harder to analyze, it offers deeper understanding. Most usability evaluations combine both types, using numbers to identify issues and user feedback to explain them.

5. Information Provided (Level):

Different evaluation techniques provide different levels of information.**Low-level information** focuses on specific interface details such as font readability, icon recognition, or menu size and is often obtained through **controlled experiments.** **High-level information f**ocuses on overall usability, usefulness, and user satisfaction, which is commonly gathered through interviews and questionnaires. The choice of method depends on the type of information needed.

6. Immediacy of Response:
This factor refers to when evaluation data is collected. **Immediate techniques**, such as Think Aloud, eye tracking, and experiments, gather information while users are performing tasks, providing fresh and accurate feedback. However, they may influence user behavior. **Delayed techniques**, such as interviews, post-task walkthroughs, and questionnaires, collect feedback after task completion, causing less interruption but risking loss of details due to memory limitations.

7. Intrusiveness:
Intrusiveness describes **how much an evaluation method interferes with normal user behavior**. Highly intrusive methods, such as Think Aloud and interviews during task performance, may alter how users behave because they are constantly aware of being evaluated. Less intrusive methods, such as system logging and distant observation, allow users to behave more naturally and produce more realistic results. Generally, methods that collect information immediately tend to be more intrusive.

8. Resources:
The choice of evaluation method is often limited by available resources such as equipment, time, money, participants, and expertise. Some techniques require expensive tools like eye trackers or sensors, while others need large numbers of participants or specialized skills in statistics and modeling. In certain situations, such as military or space systems, real users or environments may not be accessible, making simulations necessary. Resource availability often determines which evaluation methods are practical.

9.5.2 A Classification of Evaluation Techniques

Using these factors, the techniques can be classified to identify those that best fit requirements.

Analytic techniques:

Factor	Cognitive walkthrough	Heuristic	Review-based	Model-based
Stage	Throughout	Throughout	Design	Design

Style	Laboratory	Laboratory	Laboratory	Laboratory
Objective?	No	No	As source	No
Measure	Qualitative	Qualitative	As source	Qualitative
Information	Low level	High level	As source	Low level
Immediacy	N/A	N/A	As source	N/A
Intrusive?	No	No	No	No
Time	Medium	Low	Low–medium	Medium
Equipment	Low	Low	Low	Low
Expertise	High	Medium	Low	High

Experimental and query techniques:

Factor	Experiment	Interviews	Questionnaire
Stage	Throughout	Throughout	Throughout
Style	Laboratory	Lab/field	Lab/field
Objective?	Yes	No	No
Measure	Quantitative	Qual./quant.	Qual./quant.
Information	Low/high level	High level	High level
Immediacy	Yes	No	No
Intrusive?	Yes	No	No
Time	High	Low	Low
Equipment	Medium	Low	Low
Expertise	Medium	Low	Low

Observational techniques:

Factor	Think aloud	Protocol analysis	Post-task walkthrough
Stage	Implementation	Implementation	Implementation
Style	Lab/field	Lab/field	Lab/field
Objective?	No	No	No

Measure	Qualitative	Qualitative	Qualitative
Information	High/low level	High/low level	High/low level
Immediacy	Yes	Yes	No
Intrusive?	Yes	Yes (except system logs)	No
Time	High	High	Medium
Equipment	Low	High	Low
Expertise	Medium	High	Medium

Think aloud assumes a simple paper-and-pencil record; protocol analysis includes video, audio and system recording.

Monitoring techniques:

Factor	Eye tracking	Physiological measurement
Stage	Implementation	Implementation
Style	Lab	Lab
Objective?	Yes	Yes
Measure	Quantitative	Quantitative
Information	Low level	Low level
Immediacy	Yes	Yes
Intrusive?	No (if not head-mounted)	Yes
Time	Medium/high	Medium/high
Equipment	High	High
Expertise	High	High

mcqs frm random gemini chat

Q1: Your development team has built a new system and is currently in the early design phase. You have zero budget and no real users available. What should be your first evaluation step?

A) Conduct an Experimental Evaluation using a Within-Subjects design.

B) Conduct a Heuristic Evaluation to fix obvious UI issues and design violations. ✅

C) Conduct a Field Study to understand the real environment.

D) Use the GOMS model to measure typing and clicking time.

Explanation: Heuristic evaluation is best for early designs as it uses rules of thumb to find usability problems and requires no real users.

Q2: You need to evaluate your new software. According to standard practices, what is the best order of applying evaluation methods to save time and resources?

A) Model-Based -> Cognitive Walkthrough -> Heuristic Evaluation

B) Cognitive Walkthrough -> Heuristic Evaluation -> Model-Based

C) Heuristic Evaluation -> Cognitive Walkthrough -> Model-Based ✅

D) Heuristic Evaluation -> Model-Based -> Cognitive Walkthrough

Explanation: You should first apply heuristic evaluation to fix obvious UI issues, then cognitive walkthrough to improve learnability, and finally model-based evaluation to optimize performance.

Q3: You are conducting a Cognitive Walkthrough of an e-commerce app. The user needs to add an item to the 'Wishlist', but the 'Heart' icon is hidden and the user cannot find it. Which key question of the Cognitive Walkthrough does this fail?

A) Goal Match ("Is this what I want to do?")

B) Visibility ("Can I find it?") ✅

C) Recognition ("Is this the right button?")

D) Feedback ("Did it work?")

Explanation: Visibility focuses on whether the user can actually see the option or button needed to perform their task.

Q4: A user submits a form without receiving any error message, but the data does not save to the database. In a Heuristic Evaluation, what severity rating would this issue receive?

A) 1 (Cosmetic problem only)

B) 2 (Minor usability problem)

C) 3 (Major usability problem)

D) 4 (Usability catastrophe) ✅

Explanation: A severity rating of 4 indicates a usability catastrophe that makes it imperative to fix the issue before the product can be released.

Q5: You are building a messaging app and want to predict exactly how many seconds it will take a user to type and send a message. Which Model-Based method should you use?

A) GOMS Model

B) Dialog Models

C) Keystroke-Level Model (KLM) ✅

D) Design Rationale

Explanation: KLM specifically predicts the time taken for actions by measuring clicks, typing, and delays.

Q6: You are testing Dark Mode and Light Mode for your project. Group A only tests Dark Mode, and Group B only tests Light Mode. What kind of experimental design is this, and what is its main benefit?

A) Within-Subjects Design; the benefit is that fewer users are needed.

B) Between-Subjects Design; the benefit is that there is no 'Learning Effect'. ✅

C) Between-Subjects Design; the benefit is that fewer users are needed.

D) Within-Subjects Design; the benefit is that there is no 'Learning Effect'.

Explanation: In a between-subjects design, different users try different conditions, which prevents the learning effect from skewing results.

Q7: Your Null Hypothesis is: "Changing the button color has no effect on the click rate." What should be your main goal during the experiment?

A) To prove the null hypothesis true.

B) To prove the null hypothesis wrong (reject it) to show your design makes a difference. ✅

C) To ignore the null hypothesis and create a new one.

D) To only modify the dependent variable.

Explanation: The goal of experimental evaluation is usually to prove the null hypothesis wrong, demonstrating that the design changes actually had an effect.

Q8: You are testing an e-learning app in a university classroom where it is noisy, the internet keeps dropping, and friends are interrupting. What type of evaluation is this?

A) Laboratory Study

B) Field Study ✅

C) Expert Evaluation

D) Heuristic Evaluation

Explanation: Field studies are conducted in the user's real, noisy, and uncontrolled environment to observe true behavior and context.

Q9: You observed a user silently performing a task. After the task was completed, you showed them the recording and asked, "Why did you pause here for 10 seconds?". Which technique is this?

A) Think Aloud

B) Cooperative Evaluation

C) Post-Task Walkthrough ✅

D) Automated Analysis

Explanation: A post-task walkthrough involves replaying the session to the user after they finish to reveal their hidden thinking and explain unclear actions.

Q10: You collected data from 4 users, and their task completion times were 10s, 15s, 20s, and suddenly 95s. Because the data is not normally distributed (not a bell curve), which statistical test should you use?

A) Parametric Test

B) Non-Parametric Test ✅

C) Contingency Tables

D) Protocol Analysis

Explanation: Non-parametric tests are used when data is not normally distributed, using ranking instead of exact values to handle outliers safely.

Q11: Why is evaluating group systems (like a shared Google Doc) much more complex than evaluating single-user systems?

A) Because model-based techniques do not work on groups.

B) Because it only involves Human + Computer interaction.

C) Because it involves Human + Human interaction, meaning group dynamics can heavily affect the outcome. ✅

D) Because groups are always tested in a laboratory.

Explanation: Group evaluations are complex because human-to-human interaction introduces varying group dynamics, making results less predictable.

Q12: You are observing a user using an eye-tracking machine. The user's eyes are jumping everywhere and they have a very long fixation duration on random areas. What does this indicate?

A) The design is highly intuitive.

B) The user is relaxed and enjoying the system.

C) The user is confused and the design is likely poor. ✅

D) The system has a very low cognitive load.

Explanation: Long fixation durations and eyes jumping everywhere (complex scan paths) typically indicate confusion and bad design.

Q13: A user states, "This system was easy to use." However, their Heart Rate and GSR (sweat) readings indicate extreme stress. Which data should you trust more and why?

A) The user's statement, because query methods are more reliable.

B) The physiological measurements, because they provide real, objective data without bias. ✅

C) Reject both and conduct a heuristic evaluation.

D) The user's statement, because physiological devices always produce errors.

Explanation: Physiological measurements capture objective, real reactions directly from the body, bypassing subjective biases and hidden emotions.

Q14: What is considered the "Big Challenge" when conducting Protocol Analysis?

A) Users frequently forget to speak.

B) Data overload and synchronization issues between video, audio, and logs. ✅

C) Experts do not behave like real users.

D) The null hypothesis is easily proven.

Explanation: When combining multiple recording methods, the biggest challenge is data overload and the difficulty of synchronizing video, audio, and click logs.

Q15: You are directly measuring the 'Time taken to complete a task' and the 'Number of clicks'. What type of data category does this fall into?

A) Qualitative Data

B) Subjective Data

C) Quantitative Data ✅

D) Physiological Data

Explanation: Quantitative data consists of exact numbers, such as time taken, clicks, and error counts, which are easy to analyze.

Q16: Which of the following is the best example of Nielsen's heuristic 'Recognition rather than recall'?

A) Providing an "Undo" button in the system.

B) Showing suggested searches in Google as the user types. ✅

C) Displaying a progress bar for file uploads.

D) Disabling the submit button if a form is incomplete.

Explanation: Suggested searches show options directly to the user, saving them from having to retrieve or recall information from their memory.

Q17: In group experimental studies, playing the 'Desert Survival Game' where a team must rank items together falls under which task category?

A) Creative Tasks

B) Decision Games ✅

C) Conflict + Cooperation Tasks

D) Time-Critical Tasks

Explanation: Decision games require the group to collaborate and reach an agreement, such as ranking items together.

Q18: What is the primary disadvantage of using Questionnaires instead of Interviews for user evaluation?

A) They cannot reach a large audience.

B) They are highly time-consuming to administer.

C) They are less flexible, preventing you from probing deeply into user issues. ✅

D) They can only be filled out by usability experts.

Explanation: While questionnaires reach many users quickly, they are less flexible and less probing than interviews, meaning you cannot ask follow-up questions easily.

Q19: In experimental evaluation, what is the relationship between the Independent Variable (IV) and the Dependent Variable (DV)?

A) The IV is measured, and the DV is changed.

B) Both are kept at the exact same value.

C) The IV is what the experimenter changes, and the DV is the outcome that is measured. ✅

D) The IV is only used in lab studies, while the DV is used in field studies.

Explanation: The Independent Variable is manipulated (e.g., button color), and the Dependent Variable is measured to see the effect (e.g., time to click).

Q20: What is the core principle regarding the timing of Evaluation in HCI?

A) It occurs only once at the very end of system development.

B) It is a continuous process throughout the design cycle: Test -> Improve -> Test again. ✅

C) Its only purpose is to check UI aesthetics and colors.

D) It must always be done by experts without real users.

Explanation: Evaluation is not a final step; it happens throughout the design cycle as a continuous feedback loop to redesign and improve the system.